Mutali Mphephu

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Professional Summary

ECSA-registered Engineer with a Postgraduate Diploma in Electrical Engineering (University of Witwatersrand) and strong hands-on experience in C# and C++ game and systems development. Proven track record building high-performance, low-latency multiplayer experiences and cross-platform 2D/3D projects (Unity and Unreal engine), with deep focus on real-time rendering, lighting, profiling and code optimization for desktop and console. Skilled in architecting modular, data-driven frameworks, editor tooling, API integrations, and collaborating closely with developers and designers to deliver proof of concepts. Experienced with GIT, CI/CD and Linux workflows, ready to drive C# development.

Core Competencies

[Digital Signal Processing (MATLAB/Simulink)] [C/C++ (modern standards, OOP)] [Python] [C#] [Visual Scripting] [Data structures, algorithms] [Concurrency] [Code optimization] [GIT Version Control] [CI/CD (GitHub Actions, GitLab CI)] [Linux (Ubuntu, ROS)] [Networks ROS, Unity, RESTful APIs/JSON] [UI/UX (Unity)] [Embedded systems (Arduino)] [Azure deployment] [HTML/CSS/JavaScript] [Visual Studio Code] [Microsoft Office] [Agile/Scrum/Kanban] [3D Development] [GitHub/GitLab] [Click Up] [Miro , Figma, Pen pot] [Virtual Reality (Unity/Unreal)] [AutoCAD] [LTspice , Proteus, FEMM] [Power World] [Firebase] [Photon] [Vuforia] [PlayFab] [node.js] [Android , iOS deployment]

Professional Experience

Software Engineer
CSIR
08/2022 – Present

- I specialized in VR development, designing and building multiplayer experiences in Unity and Unreal Engine.
- I specialized in high-performance app development, designing and building multiplayer experiences in Unity/Unreal Engine with low-latency real-time processing and RESTful API integrations (JSON).
- Optimized real-time performance in multiplayer environments, achieving sub-50ms latency; modernized legacy codebases for cross-device stability.
- Led UI/UX design for shared experiences using Unity, Figma, and Miro; deployed via GitHub Actions, Azure, and CI/CD pipelines for mobile and desktop apps.
- Developed and deployed static/responsive websites using VS Code, GIT, and Azure hosting, ensuring security, performance profiling, and scalability.
- Integrated real-time RESTful APIs (Google Maps, weather, population data) into Unity apps, handling asynchronous processing and JSON parsing for dynamic e-commerce-like visualizations.
- Collaborated in daily Agile standups (Scrum/Kanban) with cross-functional teams (designers, PM); incorporated user feedback to refine apps and test for stability.
- Published/deployed applications to cloud stores (Azure/GitHub), mirroring App Store processes.
- Provided guidance to juniors, participated in client negotiations/proposals to drive business growth
- Actively participated in 4IR community events, workshops, and seminars to promote our work and establish industry connections.

Software Engineer
XR-Lab/RMCERI
10/2020 – 08/2022

- Created animations for 3D models and designed UI for VR applications.
- Developed VR environments in Unreal Engine and implemented user interactions using virtual scripts.
- Designed UI/UX and developed VR environments in Unreal Engine with real-time user interactions (scaling, grabbing, teleporting), optimizing algorithms for performance on resource-constrained devices.
- Utilized MATLAB/Simulink for analysis and system modeling, contributing valuable insights to project planning.
- Collaborated with team members to troubleshoot technical issues and refine VR simulations.
- Reviewed 3D modeling and simulation work to ensure quality and adherence to project requirements.
- Led VR training simulations for industrial applications (welding jigs, RGS&BPM), implementing snap-to-slot mechanics and sensor-like data processing.
- Developed welding Jig safety training simulations for industrial applications.

Junior Electrical Engineer
Aveng Grinaker-lta
01/2019 – 10/2020

- Used engineering principles to develop innovative solutions for client requirements.
- Assisted in major process plant shutdowns and routine maintenance inspections.
- Communicated effectively with team members and external stakeholders to streamline maintenance operations.
- Working with quality department to check if the work completed onsite is adhering to site specification.

Electrical Engineering In-Service Trainee (P2)
Aveng Grinaker-Ita
06/2018 – 12/2018

- Collaborated on the installation of junction boxes, conducted glanding, and terminated cables, ensuring proper electrical connections.
- Conducted site surveys and provided detailed field reports on installation progress, focusing on conduits, piping, ducting, and cabling.
- Installed plant lights and welding socket switches in adherence to zoning and classified area regulations for safety and functionality.
- Assisted the safety department with incident reports, risk assessments, and toolbox talks, prioritizing a safe working environment.
- Conducted testing of electrical systems and circuits using various tools to ensure compatibility and safety.
- Investigated on-site cable routes, marking, numbering, and installed cables for efficient and organized cable reticulation.

Electrical Engineering In-Service Trainee (P1)
RPS Switchgear
01/2018 – 06/2018

- Conducted testing and wiring of Tharisa Mine panels to ensure functionality and compliance with standards.
- Installed, commissioned, and maintained MV and LV switchgear systems for effective electrical distribution.
- Pulled underground cables, prepared bedding, and installed cable joints to facilitate efficient cable routing and connectivity.
- Educated customers on the functionality of metering systems, ensuring understanding and satisfaction.
- Attended emergency callouts, conducting testing and repair work promptly to minimize downtime and ensure operational continuity.

PROJECTS

Beyond Reality: An application of Extended Reality & Blockchain in the Metaverse Click here
In 2023, at CSIR, I participated in the Beyond Reality project, focusing on the application of Extended Reality (XR) and Blockchain in Metaverse. My responsibilities included C++/visual scripting in the Unreal Game Engine, exploring XR technologies, and integrating blockchain solutions. This project aimed to push the boundaries of reality through immersive experiences, NFTs, and blockchain integration, demanding creativity, technical expertise, and a futuristic vision which also resulted in a research paper.

Virtual reality-based training on operating the reconfigurable guillotine shear and bending press machine (RGS&BPM) Click here
In 2022, at RMCERI/XR Lab, I led a project focused on virtual reality-based training for operating the reconfigurable guillotine shear and bending press machine (RGS&BPM). This project involved developing a VR environment with functionalities such as scaling, teleporting, grabbing, and snap to slot. It required a deep understanding of VR technology, effective simulation design, and user interaction to provide immersive and realistic training experiences.

An empirical framework for developing and evaluating a Virtual Assembly Training System in learning factories Click here
At RMCERI in 2021, I contributed to the development and evaluation of a Virtual Assembly Training System in learning factories. My role involved visual scripting in the Unreal Game Engine, prototyping VR solutions for industry, and facilitating training workshops for engineering students. This project aimed to enhance manufacturing training processes using VR technology, requiring innovation, collaboration, and educational expertise.

Safety Hazard Simulation Project Click here
I contributed to a project at CSIR's EDT4IR Center, focusing on using Extended Reality (XR) technology to enhance safety training in mining and construction industries. My role involved developing a software prototype for safety hazard simulations and implementing a website/login system. I conducted research to identify effective XR-based training methods and contributed to building evidence-based guidelines for industry adoption. This project aims to improve safety practices and maximize the impact of training initiatives within the country.

Solar PV Integration in Butterfly Roofed House, Varsity Project, University of the Witwatersrand (2022): Managed the full design and testing of a solar power setup for a special butterfly-roof house. Used AutoCAD for layouts, TinkerCAD for models, and Arduino simulations to track energy flow and connect to the grid. Delivered working prototypes that showed reliable power management, building expertise in data handling and system controls.

White Line Follower Car Project, University Project: Led the development of an autonomous line-following robot using Arduino UNO and Proteus simulation software. Implemented real-time control algorithms for sensor data processing, path optimization, and motor control, achieving sub-50ms response latency through efficient data structures and concurrency handling.

EDUCATION

- **Post Graduate Diploma in Electrical Engineering**
  Witwatersrand University, 02/2022 – 12/2023
- **Bachelor of Technology in Electrical Engineering**
  Tshwane University of Technology, 07/2019 – 11/2022
- **National Diploma in Electrical Engineering**
  Tshwane University of Technology, 01/2016 – 12/2018
- **National Senior Certificate (Matric)**
  Miriyavhavha Tech Sec School, 2015

ACHIEVEMENTS

- 9th ICAT conference 3rd prize winners Certificate (2020)
- Plan A business Simulation Awareness Certificate (2021)
- National Certificate -Interactive Media NQF5
- Udemy Unity Certificates
- FNB Academy Full Stack Developer Course

REFERENCE

Available Upon Request